A Charge + Light Self-Supervised Foundation Model for MicroBooNE LArTPC Open Data

Technical Note — Round 1

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Abstract

We present a prototype multi-modal foundation model for liquid-argon TPC data that takes reconstructed charge hits and PMT light as a single token sequence. A Transformer backbone is pretrained without labels, by masked autoencoding, on 831k MicroBooNE overlay events. We then evaluate it on four downstream tasks with small task heads: hit-level and event-level neutrino/cosmic separation, flavor classification (ν_μCC/ν_eCC/NC) and neutrino-energy regression. Labeled-data fractions range from 0.1% to 100%.

Pretraining clearly helps when labels are scarce. With 0.1% labels, the fine-tuned charge-plus-light model beats training from scratch on every task; for example, hit-level F1 is 0.77 versus 0.22.

Adding light improves every task and is decisive for neutrino/cosmic separation: event-level AUC is 0.97 versus 0.65 for charge only.

At fixed compute, a 6.6M-parameter model saturates at about 10% of the pretraining data. A 21M-parameter model trained twice as long improves all probes, and with it more pretraining data helps consistently. On neutrino/cosmic separation the best models clearly beat the Pandora and flash-PE reference baselines: hit F1 0.95 versus 0.63, and event AUC 0.999 versus 0.82 and 0.91.

We also show that, with fixed-grid charge patches, the masked-reconstruction target is not learnable unless each patch is positioned at its charge centroid. Round 1 used about 29 Polaris node-hours.

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*Prepared with the assistance of Claude (Anthropic). Code: `https://github.com/HaiwangYu/fm4npp-lartpc`, branch `round1-prototype`, package `multi-modal-qlmatch/qlfm`. Review thread: GitHub issue #1.

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1 Introduction and goals

Foundation models (FMs) learn reusable representations from unlabeled data. Small task-specific heads are then trained on top, as in language and vision. Recent neutrino and nuclear-physics work applies this idea to 3D charge point clouds [6, 9, 10] and to collider TPC space-points [8].

A LArTPC records two correlated measurements of the same event:

- ionization charge on the wire planes;
- scintillation light on the PMTs.

Classical reconstruction relates the two explicitly, for example through Wire-Cell's many-to-many charge–light matching [2, 3]. This round tests whether a single self-supervised backbone can learn these correlations, and the physics they carry, directly from paired charge and light.

Round 1 asks three questions, using the MicroBooNE public open samples [1]:

Q1 Does SSL pretraining help? We compare a downstream model trained from scratch with a frozen FM plus head and with a fine-tuned FM, across label budgets.

Q2 Does light help? We compare charge-only (C) with charge+light (CL) inputs.

Q3 How does the FM scale with the amount of pretraining data (1%, 10%, 50%, 90%)?

The compute allowance was 100 node-hours on the Polaris `debug`/`debug-scaling` queues, which have a 1-hour walltime. Section 6 gives the stopping criterion.

2 Data and tasks

2.1 Samples, preprocessing and splits

We use two NoWire HDF5 samples:

- the BNB inclusive overlay (Zenodo 8370883): 753,467 events, 195 GB;
- the ν_e-CC overlay (Zenodo 7261921): 18 of 20 files, 77,340 events.

Together they give 830,807 events. Both samples are GENIE+Geant4 neutrinos overlaid on real off-beam cosmic data, so cosmic hits carry no MC truth.

The HDF5 tables are converted into a column-wise, memory-mappable cache (146 GB for the inclusive sample):

- **per hit:** plane, wire, time, integral, rms; truth (deposited neutrino energy and category); and a Pandora ν-slice flag, used only as a baseline;
- **per optical hit:** channel, time and PE;
- **per event:** truth, the Pandora score and the largest beam flash.

Splits are assigned by a hash of (run, subrun). Overlays that share an off-beam cosmic frame therefore stay in one partition. The pretraining subsets (1/10/50/90% of all events) and the label pools (0.1/1/10/100% of train) are nested random subsets of the train partition.

Table 1: Partitions of the merged corpus. Following the review decision, the ν_e sample is mixed into every partition.

Split	Events	ν_e-sample share
train	747,068	9.4%
val	41,912	9.2%
test	41,827	7.2%

Audit (inclusive sample).

- 15.4k hits per event (median).
- 44% of events have the ν vertex in the active volume (AV).
- In 21% of events the neutrino leaves ≤ 5 hits.
- The beam flash follows the neutrino interaction time by 0.478 μs (5–95%: 0.457–0.507 μs).
- About 1% of optical hits fall outside the 23.4 μs beam window and are dropped.

2.2 Downstream tasks

Table 2 summarizes the tasks. All definitions live in `qlfm/data/labels.py`.

Table 2: Downstream task definitions. AV = active volume; FV = fiducial volume, 3 cm inside the AV; "visible" means ≥ 30 collection-plane ν hits.

Task	Definition	Train counts	Metric
T1 hit ν/cosmic	Per hit: ν if the MC ν-deposit exceeds 0.05 MeV. Evaluated on events with a visible in-AV ν	$\sim$6% ν hits	charge-wtd F1
T2 event ν/cosmic	0 = no ν charge ($\leq$5 ν hits); 1 = visible in-AV ν; 2 = out-of-AV ν leaving >5 hits	142k/333k/241k	AUC (1 vs 0)
T3 flavor	FV + visible: ν_μCC / ν_eCC / NC	200k/68k/49k	macro-F1
T4 energy	FV + visible + CC; regress $\log E_\nu$	268k	resolution†

† Half the 16–84% interval of $(E_{\text{pred}} - E_{\text{true}})/E_{\text{true}}$.

Reference baselines.

- **T1:** Pandora hits in the slice with the highest `nu_score`.
- **T2:** Pandora's maximum `nu_score` gives AUC 0.82; the beam-flash PE in 3–6 μs gives AUC 0.91.

3 Model

Figure 1 shows the architecture.

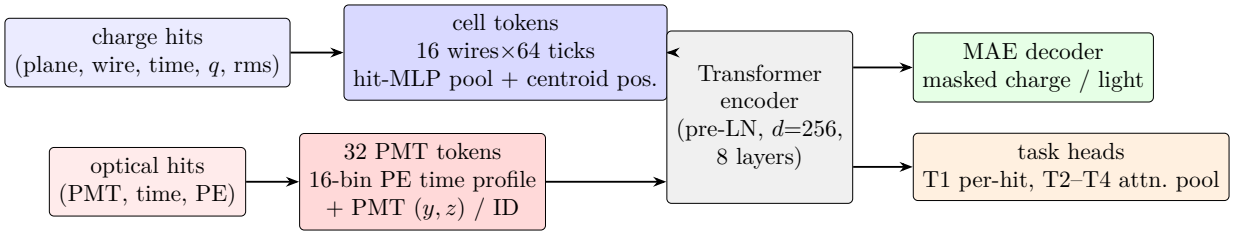


Figure 1: Architecture. Charge cells and PMTs become 256-dimensional tokens in a shared space, with a modality embedding added to each. Pretraining uses the MAE decoder; downstream tasks attach small heads to the encoder.

3.1 Tokenization

Charge tokens.

- Each wire plane is divided into 16-wire × 64-tick cells, about $4.8 \times 3.5\,\mathrm{cm}^2$. Every occupied cell becomes one token.
- This gives about 2.0k tokens per event (median; 99th percentile 3.4k; capped at 4096).
- A shared per-hit MLP takes each hit's relative wire, relative time, $\log q$ and rms. Its outputs are max- and mean-pooled over the cell and concatenated with the cell's hit count and total charge.
- The token's position is its charge-weighted centroid in physical units (drift x and wire coordinate, in cm). It is encoded by 10-octave Fourier features, plus an embedding of the centroid's offset inside the cell, plus a plane embedding.

Light tokens.

- There are 32 tokens per event, one per PMT.
- Each carries $\log(1 + \mathrm{PE})$ in 16 time bins spanning 0–24 μs, binned more finely around the expected neutrino flash, plus the total PE.
- The position is encoded by Fourier features of the PMT (y, z) plus a PMT-ID embedding.

A modality embedding is added to every token. The charge-only model simply has no light tokens. Tokenization runs on the GPU from raw hits.

3.2 Walk-through on one event

Figures 2–4 follow one event from the raw data to the token vectors. The event is run 6182, subrun 91, event 4562: a 1.6 GeV ν_eCC shower at $z \approx 910$ cm, overlaid on cosmic rays. Table 3 lists the tensor shapes at each step. All plots are made with the model's own tokenizer (`scripts/r1_io_figures.py`).

Table 3: Tensor shapes for this event ($d = 256$ for the 6.6M model).

Step	Charge	Light
raw input	11,469 hits × (plane, wire, tick, q, rms)	390 optical hits × (PMT, time, PE)
per-element features	11,469 × 4 (in-cell Δw, Δt, $\log q$, rms)	–
tokens	1,502 cells	32 PMTs × 17 (16 time bins + total)
token vectors	1,502 × d	32 × d
encoder input / output	one sequence of 1,534 vectors × d; every token attends to every other	
MAE target (masked)	34 per cell (4×8 sub-grid + 2 summaries)	17 per PMT

Charge input of one event (run 6182/91/4562; true νe CC, 1.62 GeV, red box = neutrino). Top: reconstructed wire hits. Bottom: one token per occupied 16-wire × 64-tick cell (1,502 tokens in total).

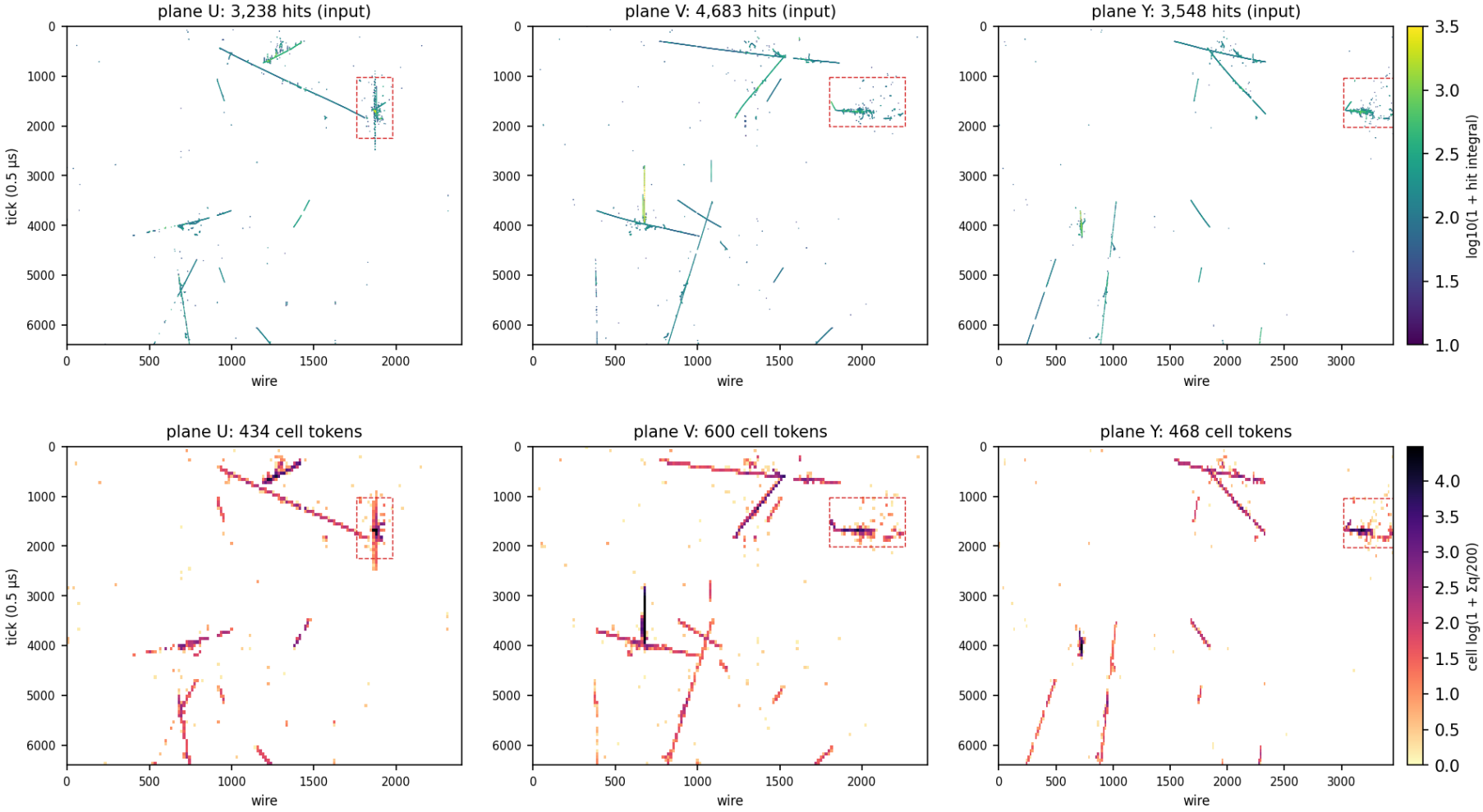
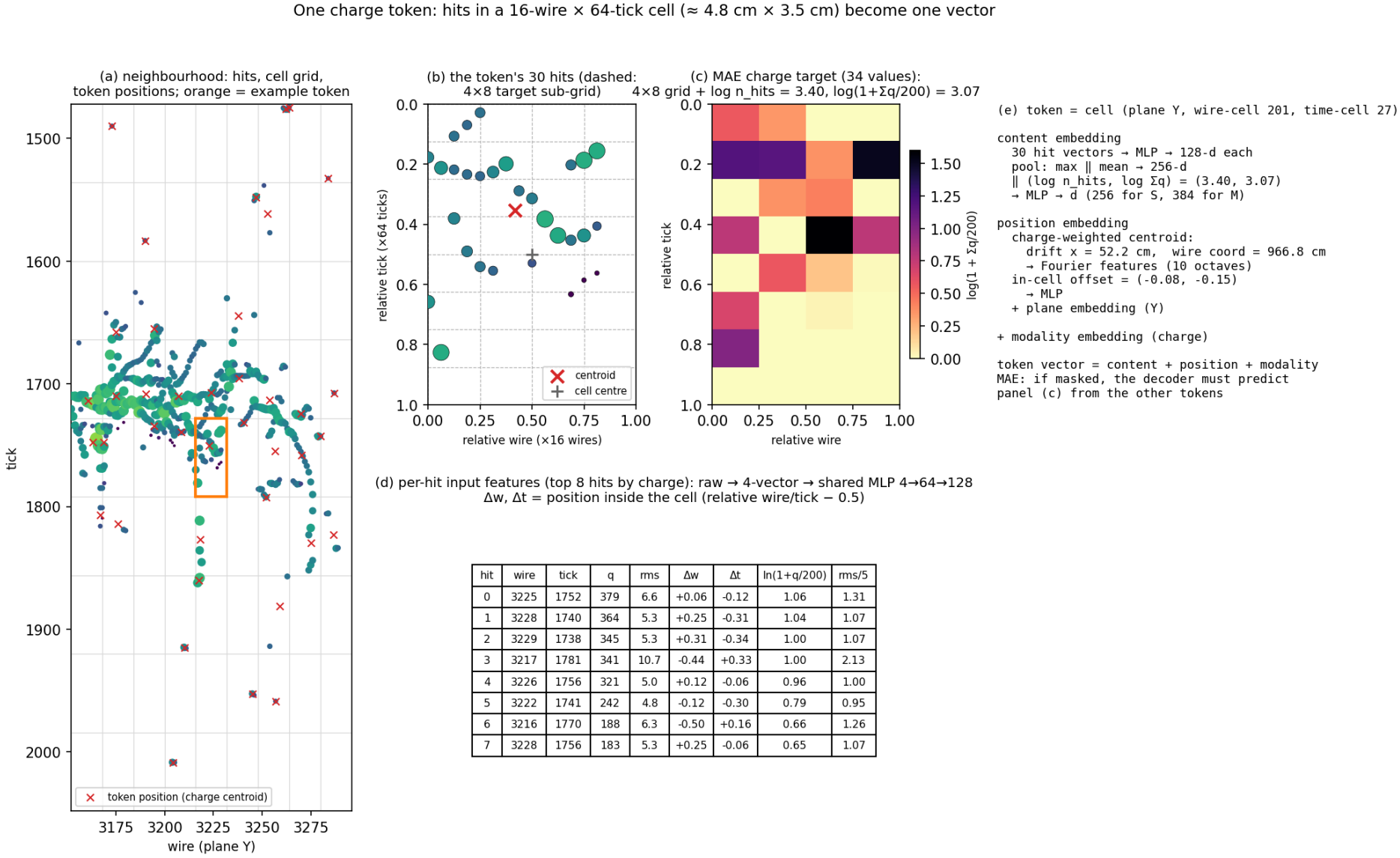


Figure 2: Charge input. Top: the reconstructed wire hits of the three planes (wire vs. drift time). Bottom: the tokens. Every occupied 16-wire × 64-tick cell becomes one token, coloured by its total charge. The dashed box marks the neutrino; everything else is cosmic.



hit	wire	tick	q	rms	Δw	Δt	ln(1+q/200)	rms/5
0	3225	1752	379	6.6	+0.06	-0.12	1.06	1.31
1	3228	1740	364	5.3	+0.25	-0.31	1.04	1.07
2	3229	1738	345	5.3	+0.31	-0.34	1.00	1.07
3	3217	1781	341	10.7	-0.44	+0.33	1.00	2.13
4	3226	1756	321	5.0	+0.12	-0.06	0.96	1.00
5	3222	1741	242	4.8	-0.12	-0.30	0.79	0.95
6	3216	1770	188	6.3	-0.50	+0.16	0.66	1.26
7	3228	1756	183	5.3	+0.25	-0.06	0.65	1.07

Figure 3: One charge token. (a) A 9 × 9-cell neighbourhood of the neutrino shower on the Y plane. Red crosses are token positions, i.e. the charge centroids; the orange box is the example token. (b) The token's 30 hits in cell coordinates. The centroid (red) lies on the charge, not at the cell centre (grey). (c) The 34-value MAE target of this token: the charge in a 4 × 8 sub-grid of the cell, plus log hit count and log total charge. (d) The 4 features of each hit, which a shared MLP maps to a 128-d hit embedding. (e) How the token vector is assembled: content (pooled hit embeddings), position (centroid, in-cell offset, plane) and modality.

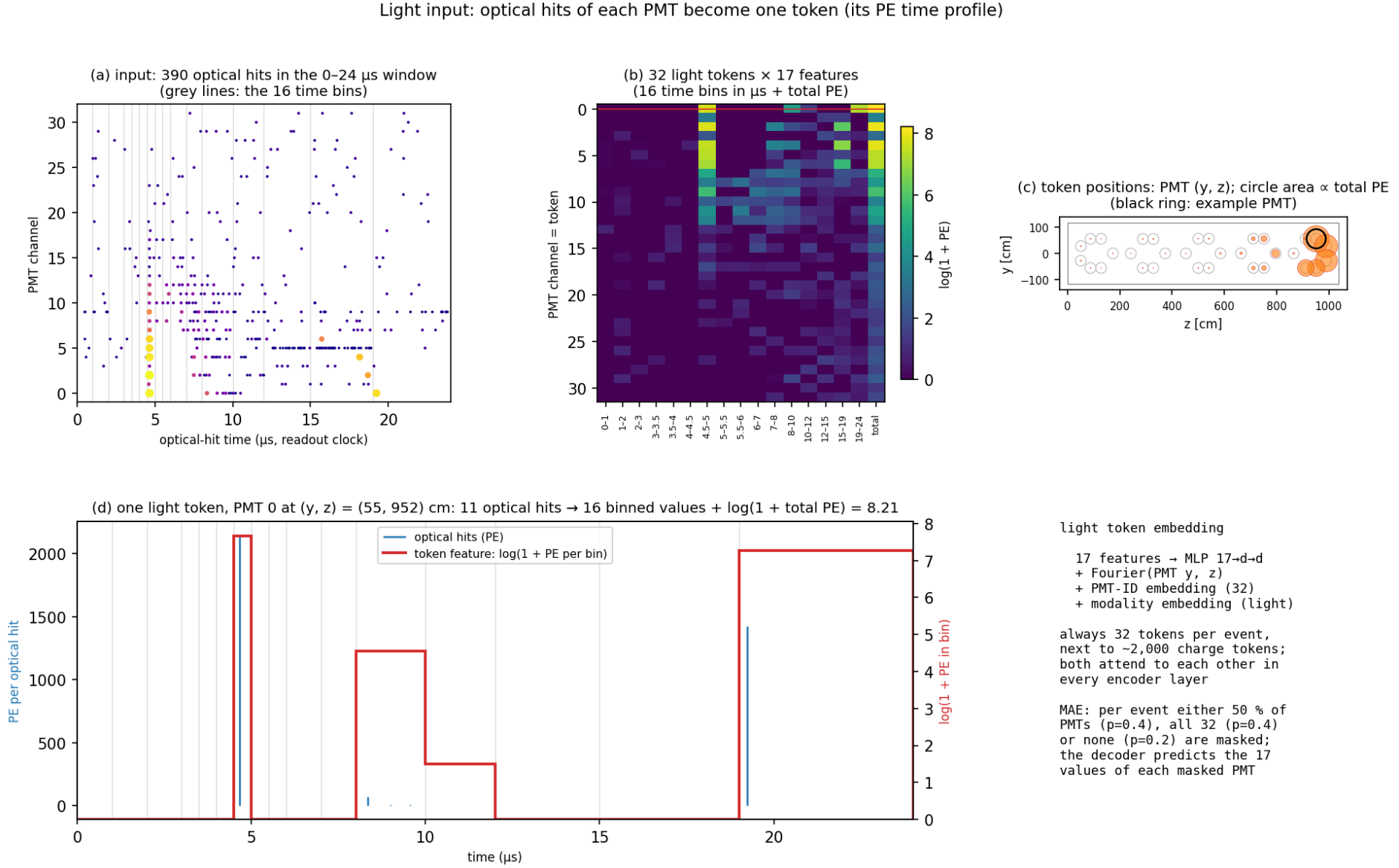


Figure 4: Light input. (a) All optical hits in the 24 μs readout window. Grey lines show the 16 time bins, which are finer around the beam flash at 3–6 μs. (b) The 32 light tokens. Each row is one PMT: $\log(1 + \mathrm{PE})$ in each time bin, plus the total. (c) Token positions, i.e. the PMT (y, z) coordinates, with circle area proportional to total PE. The beam flash sits above the neutrino at $z \approx 950$ cm. (d) One light token in detail (PMT 0). Its optical hits (blue) are binned into the 17-value feature vector (red step line). The later hit at 19 μs belongs to a cosmic flash.

3.3 Backbone

The encoder is a pre-LN Transformer [13]: 8 layers, token width d=256, 8 heads of dimension 32, and MLP hidden size 1024. It has 6.6M parameters. Pre-LN applies LayerNorm to the input of each sublayer and leaves the residual path untouched:

$$x \leftarrow x + \mathrm{Attn}(\mathrm{LN}(x)), \qquad x \leftarrow x + \mathrm{MLP}(\mathrm{LN}(x)).$$

This trains more stably than the original post-LN form. Downstream heads read the encoder output without a final LayerNorm, following [6]. Events are packed into variable-length sequences and processed with FlashAttention-2 [14].

3.4 Pretraining objective

We use a multi-modal masked autoencoder [4, 7].

Charge.

- 60% of the charge tokens are masked.
- Half of the masked budget comes from 256-tick drift-time strips masked jointly on all three planes, so the model cannot copy a hidden region from another plane's view.
- The decoder (d=192, 3 layers) predicts, for each masked cell, a 4×8 in-cell charge sub-grid plus the cell's log hit count and log total charge. Each target dimension is standardized per batch.

Light. Each event uses one of three masking modes:

- 50% of the PMTs are masked, with probability 0.4;
- all 32 PMTs are masked, with probability 0.4. This makes the task *charge→light* prediction, a learned stand-in for a photon library combined with flash matching;
- nothing is masked, with probability 0.2.

Input and output on one event. Figure 5 applies the trained 6.6M charge+light MAE to the walk-through event.

- 60% of the charge tokens are hidden from the encoder: drift-time strips across all planes plus random cells.
- All 32 PMT tokens are hidden.
- The decoder then predicts each hidden cell's in-cell charge pattern and summaries, and the full light pattern, from the visible charge alone.

Schedule. Training runs for 17k steps at a global batch of 256 events, on one Polaris node (4×A100) in about 45 min. We use AdamW with learning rate 1.5×10^{-4}, 5% warmup, cosine decay and bf16. The step count is the same for every pretraining fraction.

3.5 A fixed-grid MAE target needs patch centres

In the first version, each charge token was positioned at its cell centre. The masked-charge loss never beat a per-dimension mean predictor, while the light reconstruction learned normally.

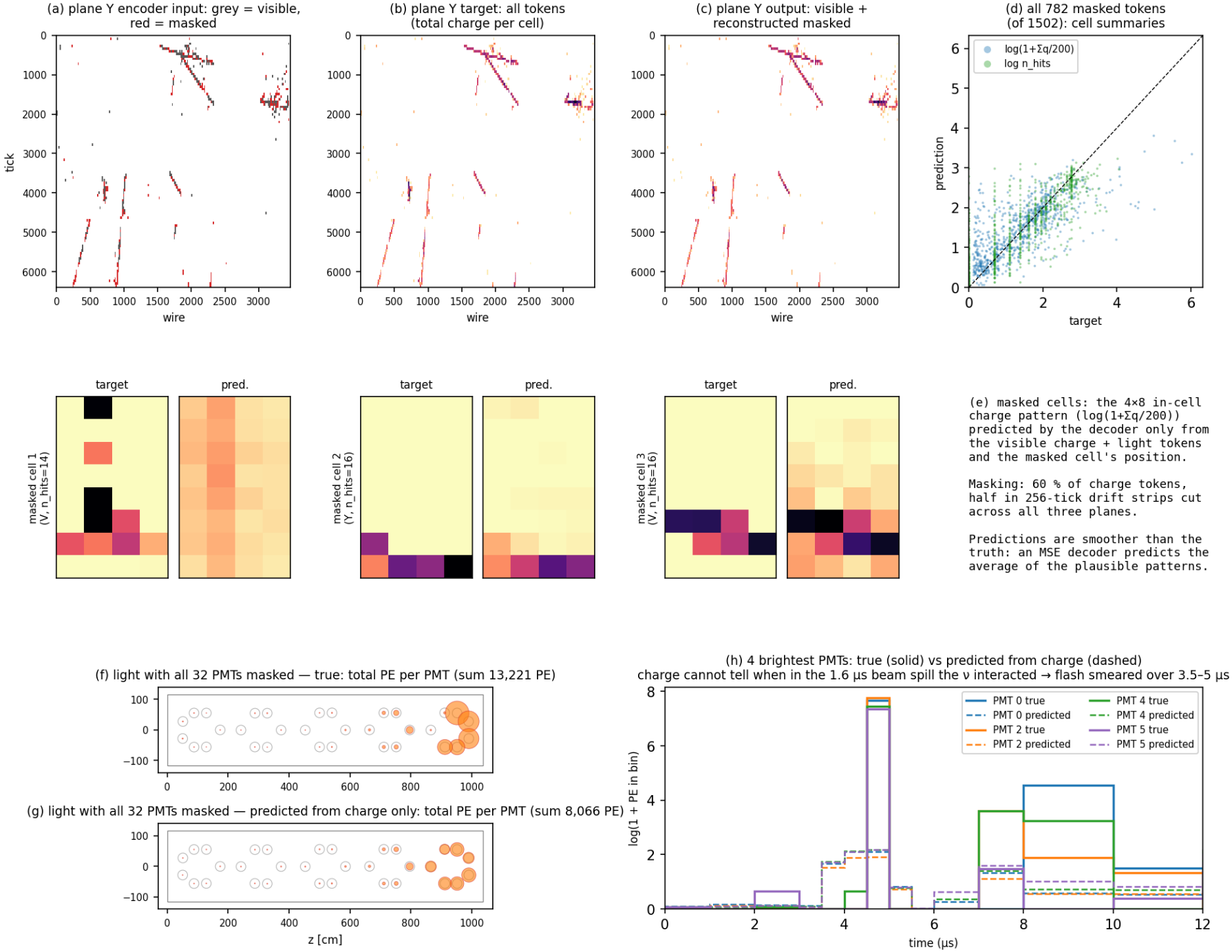


Figure 5: Masked-autoencoder input and output for the event of Sec. 3.2 (qlfm-S CL). (a) Encoder input on the Y plane: visible tokens; masked tokens are shown in red. (b) Target. (c) Output: visible tokens plus the decoder's reconstruction of the masked ones (total charge). (d) Reconstructed vs. true cell summaries for all masked tokens of the event. (e) Three masked cells: true and predicted 4×8 in-cell charge. (f, g) With all PMTs masked, the light is predicted from charge alone: true vs. predicted total PE per PMT. The *where* (the downstream PMTs above the shower) and the order of magnitude are reproduced. (h) Time profiles

Diagnosis.

- A single-batch overfit test and a target-leak test confirmed that the code was wired correctly.
- Fourier, learned-grid and combined position encodings all failed in the same way.
- A non-neural baseline settled the question. Predicting a masked cell's (hit count, total charge) as the average over its visible neighbours is *worse* than predicting the global mean, for 8×32, 16×64 and 32×128 cells.

Tracks cross a fixed grid at arbitrary offsets: a corner clip gives 1–2 hits, while a central crossing gives about 16. Unless the model knows where the track passes through the cell, the cell's content is not predictable.

Fix. Following Point-MAE [5] and PoLAr-MAE [6], tokens are positioned at their charge centroid, together with the centroid's offset inside the cell. With this change the held-out masked-charge error drops from 0.97–0.99 to 0.61× the mean baseline.

3.6 Heads and adaptation modes

Heads.

- **Event-level tasks (T2–T4):** an attentive-pooling head, in which 4 learned queries cross-attend to all tokens, followed by an MLP. It has about 0.5M parameters.
- **Hit-level task (T1):** a per-hit MLP that takes the hit embedding concatenated with the contextual feature of its token.

Modes.

- *scratch*: random initialization; everything is trained.
- *frozen*: the backbone is fixed and only the head is trained.
- *FT*: everything is trained, with layer-wise learning-rate decay 0.75.

Training and evaluation.

- Each run uses one GPU, batch size 32, and $\mathrm{clip}(30\ \text{epochs}, 400, 15000)$ steps.
- Early stopping uses a random validation subset.
- Tests use a random test subset of up to 20k events (3k for T1).
- Label budgets $\leq$1% are run with three seeds.

4 Experiments

5 Results

5.1 Label efficiency and the value of light (E3)

Best model (FT-CL, 100% labels).

- **T2:** AUC 0.999. Signal efficiency is 0.997 at 95% background rejection and 0.973 at 99%. The in-AV versus out-of-AV AUC is 0.96, and the 3-class macro-F1 is 0.82.
- **T3:** ν_μCC efficiency/purity 0.88/0.92; ν_eCC 0.80/0.72 (AUC 0.96); NC 0.65/0.63.
- **T4:** resolution 23.7%, bias -2.5%.
- **T1:** charge-weighted efficiency/purity 0.948/0.953. Pandora reaches 0.78/0.56 on the same events.

Table 4: Experiment overview. Total compute was about 29 node-hours (`sbank`: 29.6 charged on suballocation `debug`). Jobs were submitted by a queue-aware feeder; several single-GPU runs shared each node through a lock-based task queue.

ID	Content	Runs
E1	Scratch sanity checks of the label pipelines (inclusive-only corpus)	4
E2	Pretraining of FM-C and FM-CL on 90% of the data	2
E3	{scratch, frozen, FT} × {C, CL} × {0.1, 1, 10, 100}% labels × 4 tasks; 3 seeds at ≤1%	192
E4	Pretraining of FM-C and FM-CL on 1/10/50% of the data; frozen probes at 1% and 10% labels	6 + 48
E5	Checks: learning-rate scan for scratch-C; layer decay 0.9 for hit fine-tuning	8
E6	Larger model (21M parameters, 2× steps), FM-CL on 10% and 90%; frozen and FT probes	2 + 24

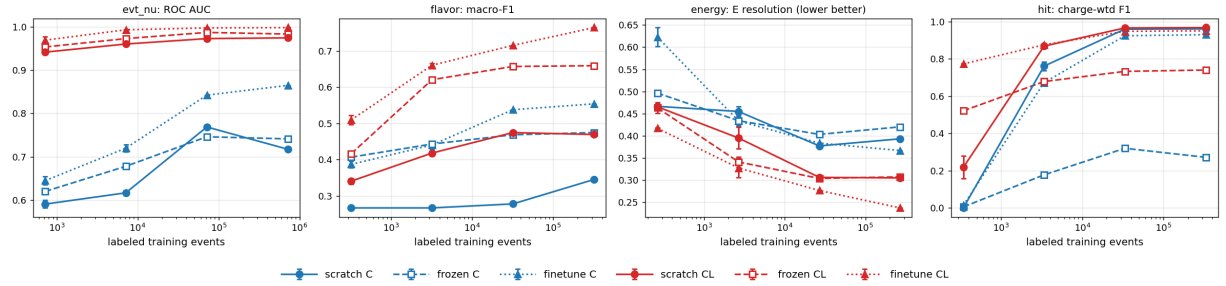


Figure 6: Test metrics versus the number of labeled training events. Colour: input (blue C, red CL). Marker and line: mode (scratch ○ solid, frozen □ dashed, FT △ dotted). Error bars show the seed spread at ≤1% labels.

Interpretation.

- **Q1: pretraining helps most with few labels.** The advantage shrinks as labels grow.
  - On T1, scratch catches up by 1% labels and is slightly ahead at ≥10%.
  - On the event-level tasks, FT-CL stays ahead even at 100%. Scratch is limited there by the downstream budget (≤15k steps, less than one epoch), so part of that gap is undertraining.
- **The frozen CL backbone is a strong feature extractor for event-level tasks.** It matches or beats scratch at every budget on T2–T4.
- **Frozen features are weak for per-hit labels.** The per-hit embedding receives no direct pretraining signal.
- **Q2: light dominates neutrino/cosmic separation.**
  - Charge-only models must find a small interaction among roughly 26 cosmic muons.
  - The in-time PMT pattern points directly to the neutrino. This is the learned counterpart of charge–light matching.
  - Light also improves flavor and energy.
- **Charge-only scratch fails on event-level tasks** at every tested learning rate (E5). Fine-tuning the same architecture from the C foundation model succeeds: flavor 0.54–0.55 and T2 AUC 0.84–0.87 at 10–100% labels.
- **No sample-origin shortcut is seen for flavor.** The ν_eCC efficiency on inclusive-sample ν_e events (n=79) is equal to or higher than on ν_e-sample events: 0.86 versus 0.80 for FT-CL.

Table 5: E3 test metrics at 0.1 / 1 / 10 / 100% labels. Entries at ≤1% are the mean of three seeds; the seed spread is typically ≤ 0.01. For T4, lower is better.

Task (metric)	Mode	C	CL
T2 event ν (AUC)	scratch	0.59 / 0.62 / 0.77 / 0.72	0.94 / 0.96 / 0.97 / 0.98
	frozen	0.62 / 0.68 / 0.75 / 0.74	0.95 / 0.97 / 0.99 / 0.98
	FT	0.65 / 0.72 / 0.84 / 0.87	**0.97 / 0.99 / 1.00 / 1.00**
T3 flavor (macro-F1)	scratch	0.27 / 0.27 / 0.28 / 0.35	0.34 / 0.42 / 0.48 / 0.47
	frozen	0.41 / 0.44 / 0.47 / 0.47	0.42 / 0.62 / 0.66 / 0.66
	FT	0.39 / 0.44 / 0.54 / 0.55	**0.51 / 0.66 / 0.72 / 0.77**
T4 energy (resolution)	scratch	0.47 / 0.46 / 0.38 / 0.39	0.47 / 0.40 / 0.31 / 0.31
	frozen	0.50 / 0.43 / 0.40 / 0.42	0.46 / 0.34 / 0.30 / 0.31
	FT	0.62 / 0.43 / 0.38 / 0.37	**0.42 / 0.33 / 0.28 / 0.24**
T1 hit (charge-wtd F1)	scratch	0.00 / 0.76 / 0.96 / 0.96	0.22 / 0.87 / **0.97 / 0.97**
	frozen	0.01 / 0.18 / 0.32 / 0.27	0.52 / 0.68 / 0.73 / 0.74
	FT	0.01 / 0.67 / 0.92 / 0.93	**0.77 / 0.88** / 0.95 / 0.95

5.2 Pretraining-data scaling (E4, E6)

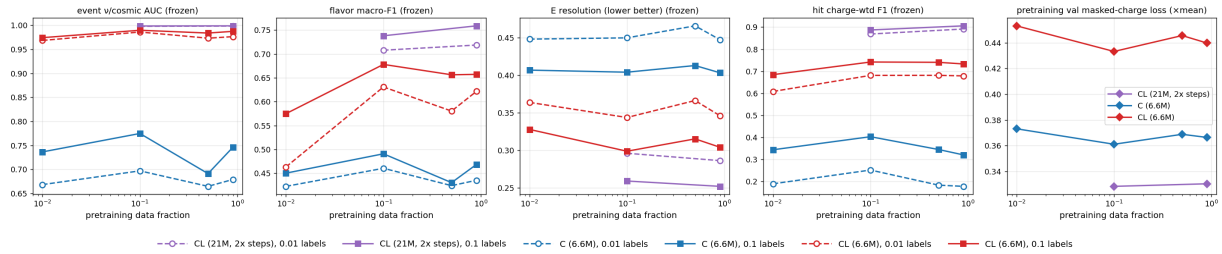


Figure 7: Frozen-probe test metrics and pretraining validation masked-charge loss versus the pretraining-data fraction. Red and blue: 6.6M CL and C models at fixed compute (17k steps). Purple: 21M CL model at 2× steps. Open circles with dashed lines: 1% labels; filled squares with solid lines: 10% labels.

Small model at fixed compute (E4).

- CL improves from 1% to 10% of the pretraining data. At 1% labels: flavor 0.46 → 0.63, T1 0.61 → 0.68, T4 0.36 → 0.34, T2 0.968 → 0.986.
- Above 10% the probes are flat within single-seed jitter of ±0.02–0.04. C is flat throughout.
- The pretraining validation loss has the same shape: CL 0.453, 0.433, 0.446, 0.440; C 0.373, 0.361, 0.369, 0.366.
- Every run sees the same 4.4M events. With 10% of the data (83k events, about 53 epochs), the 6.6M-parameter model already saturates.

Larger model (E6). We pretrain a CL model with d=384, 12 layers and 6 heads (21M parameters) for 34k steps (about 8.7M events) on 10% and on 90% of the data.

What E6 shows.

- **Capacity and compute dominate at this scale.** Going from S to M improves every probe: frozen T1 F1 by +0.15 to +0.19, frozen T3 macro-F1 by +0.08 to +0.10, and T4 resolution by about −0.05.

Table 6: Pretraining validation loss, in units of the mean-predictor baseline.

Model	Data	Masked charge	Masked light	Charge→light
S (6.6M)	10%	0.433	0.448	0.559
S (6.6M)	90%	0.440	0.446	0.556
M (21M, 2× steps)	10%	0.328	0.431	0.539
M (21M, 2× steps)	90%	0.330	0.413	0.512

Table 7: CL probe test metrics for the small (S) and large (M) models at 10% and 90% pretraining data. Bold marks where the 90% model beats the 10% model.

Probe	S 10%	S 90%	M 10%	M 90%
T2 AUC, frozen, 1% / 10% labels	0.986 / 0.990	0.976 / 0.987	0.998 / 0.999	0.998 / 0.999
T2 AUC, FT, 1% labels	—	0.993	0.998	0.999
T3 macro-F1, frozen, 1% / 10%	0.631 / 0.678	0.622 / 0.657	0.708 / 0.738	**0.719 / 0.759**
T3 macro-F1, FT, 1%	—	0.663	0.724	**0.741**
T4 resolution, frozen, 1% / 10%	0.344 / 0.299	0.346 / 0.304	0.296 / 0.259	**0.286 / 0.252**
T4 resolution, FT, 1%	—	0.307	0.292	**0.274**
T1 F1, frozen, 1% / 10%	0.681 / 0.742	0.678 / 0.733	0.869 / 0.887	**0.891 / 0.906**
T1 F1, FT, 1%	—	0.875	0.925	**0.943**

- **Frozen M features rival a fully trained small model.** At 10% labels, frozen M reaches T3 0.759 and T4 0.252. The small model fine-tuned on all labels reaches 0.765 and 0.237.
- **More pretraining data now helps.** With M, 90% beats 10% in all 9 non-saturated comparisons (T3 +0.011 to +0.021; T4 −0.007 to −0.018; T1 +0.018 to +0.022). With S, the same comparisons were flat or reversed.
- **The data benefit shows up in the light objectives.** Charge→light improves from 0.539 to 0.512, while the masked-charge loss is unchanged. The extra data mainly sharpens the learned charge–light correspondence.
- These are single seeds. The evidence is the consistent sign across all 9 comparisons, not the size of any single difference.

5.3 Checks (E5)

- **Scratch-C learning rate.** At 10% labels, learning rates 10^{-4} and 10^{-3} give flavor F1 0.277 and 0.267 and T2 AUC 0.69 and 0.56. The default 3×10^{-4} gives 0.277 and 0.77. The scratch failure is not a learning-rate artifact.
- **Hit fine-tuning with layer decay 0.9** (default 0.75).
  - At 10% labels: C 0.952 (was 0.924) and CL 0.960 (was 0.948), level with scratch (0.959/0.966).
  - At 0.1% labels: CL 0.788 (was 0.773).
  - A decay of 0.9 is recommended for dense tasks.

6 Conclusion of round 1

The round was to stop once the performance was decent and the three comparisons made physical sense, or once further iteration stopped improving the results. That point has been reached:

1. Pretraining clearly beats scratch in the low-label regime, and the advantage narrows as labels grow. For per-hit labels it disappears by about 1% labels, once fine-tuning uses a sensible layer decay.
2. Charge+light is better than charge only on every task, and decisively so for neutrino/cosmic separation. The learned charge→light prediction is non-trivial.
3. Pretraining-data scaling is limited by compute and capacity for the 6.6M model. It reappears consistently with a larger model trained for longer.
4. On neutrino/cosmic separation, the best models clearly beat the Pandora and flash-PE reference baselines.

Further gains are expected from scale, from the pretraining objective and from hit-level design. These are round-2 topics (Section 8).

7 Limitations

- **Overlay data only.** There is no beam-on or beam-off sample. The T2 negatives are overlay events in which the neutrino left no charge.
- **Beam-window light only** (23.4 μs). This is not Wire-Cell's full 4.8 ms charge–light matching.
- **Single seeds** at 10% and 100% labels and in E4/E6. Differences below about 0.02 are not significant on their own.
- **Fixed downstream budget.** At large label budgets, scratch is not trained to convergence.
- **Small models and budget.** The models (6.6M and 21M parameters) and the few-million-event pretraining budget were set by the 100 node-hour allowance.
- **Non-natural T3 class proportions.** The ν_e sample is mixed into all splits.
- **Operational label thresholds.** For example, the hit-level ν threshold of 0.05 MeV.

8 Next steps

1. **Hit-level representations.** Add a per-hit pretraining signal: hit-level masked prediction or dense self-distillation [9, 11]. Also try finer cells.
2. **Latent self-distillation objective.** An iBOT/JEPA-style objective [12] compared with MAE at matched compute.
3. **Scaling.** Grow model size and compute together, with more seeds, and add the 1% and 50% points for the M model.
4. **Backbone ablation.** FM4NPP-style per-hit tokens with a Mamba2 backbone [8].
5. **Physics realism.** Beam-off samples; interaction-level (slice) heads; a comparison with Wire-Cell's generic neutrino selection [3] at matched efficiency.

Reproducibility

- **Code:** `https://github.com/HaiwangYu/fm4npp-lartpc`, branch `round1-prototype`.
- **Main entry points:** `qlfm.data.convert`, `scripts/make_splits.py`, `qlfm.train_pretrain`, `qlfm.train_downstream`, `scripts/make_tasks.py`, `scripts/feeder.sh`, `scripts/summarize.py` and `scripts/plot_scaling.py`.

- **Outputs:** on Polaris under `/lus/eagle/projects/neutrinoGPU/yuhw/fm4npp-lartpc-work/`.
- **Companion documents:** a Markdown version of this note with the same content (`docs/technote-round1.md`) and a working log of decisions and findings (`docs/round1-log.md`).

References

[1] G. Cerati (for MicroBooNE), *MicroBooNE public data sets: a collaborative tool for LArTPC software development*, arXiv:2309.15362; data DOIs 10.5281/zenodo.8370883 and 10.5281/zenodo.7261921.

[2] MicroBooNE Collaboration, *Neutrino event selection in the MicroBooNE LArTPC using Wire-Cell 3D imaging, clustering, and charge-light matching*, JINST 16 (2021) P06043, arXiv:2011.01375.

[3] MicroBooNE Collaboration, *Cosmic ray background rejection with Wire-Cell LArTPC event reconstruction in the MicroBooNE detector*, arXiv:2101.05076.

[4] K. He et al., *Masked autoencoders are scalable vision learners*, arXiv:2111.06377.

[5] Y. Pang et al., *Masked autoencoders for point cloud self-supervised learning*, arXiv:2203.06604.

[6] *Particle trajectory representation learning with masked point modeling* (PoLAr-MAE), arXiv:2502.02558.

[7] R. Bachmann et al., *MultiMAE: multi-modal multi-task masked autoencoders*, arXiv:2204.01678.

[8] *FM4NPP: a scaling foundation model for nuclear and particle physics*, arXiv:2508.14087.

[9] *Panda: self-distillation of reusable sensor-level representations for high energy physics*, arXiv:2512.01324.

[10] *Panda Diplomacy: foundation model pre-training across particle imaging detectors*, arXiv:2609.00611.

[11] X. Wu et al., *Sonata: self-supervised learning of reliable point representations*, arXiv:2503.16429.

[12] M. Assran et al., *Self-supervised learning from images with a joint-embedding predictive architecture*, arXiv:2301.08243.

[13] R. Xiong et al., *On layer normalization in the Transformer architecture*, arXiv:2002.04745.

[14] T. Dao, *FlashAttention-2: faster attention with better parallelism and work partitioning*, arXiv:2307.08691.